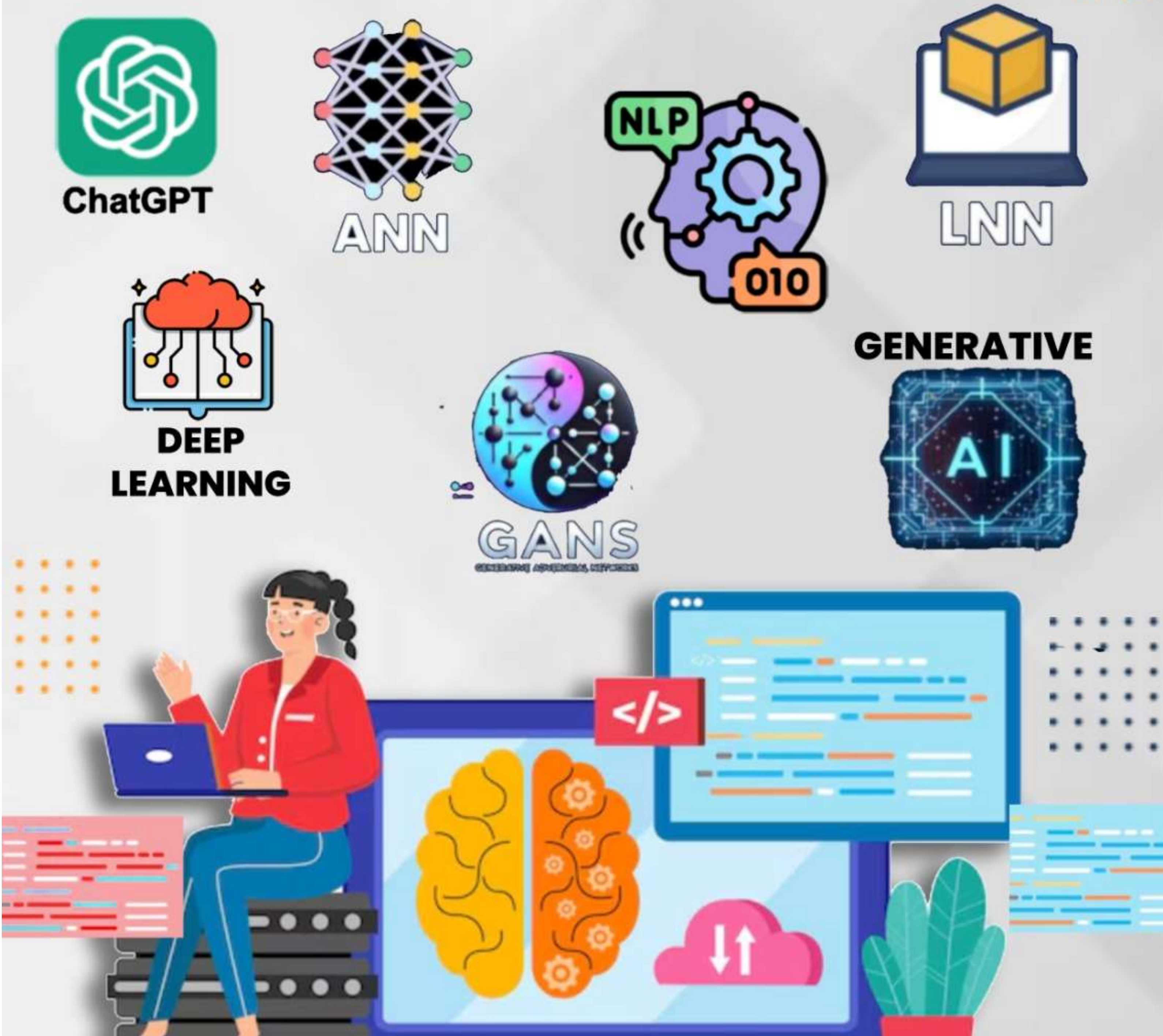
UpFlairs
...because winning takes it all

NLP , DEEP LEARNING & CHAT GPT



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A Brief About UPFLAIRS

UpFlairs is an innovative **educational technology** company with a clear mission to elevate the skills and employability of students throughout India.

Our dedicated team is committed to fostering the next generation of tech talent, equipping them with cutting-edge skills in emerging technologies and has educated more than 47K students all over the globe including IITs, NITs, Deemed Universities and other colleges.

We offer the courses that are most trending technologies of the recent era in the fields like **AI/ML, Data Science, Cloud Computing, DevOps, Full Stack Web Development, Embedded Systems, IoT, and Robotics.** These courses are meticulously designed to provide students with the practical skills and knowledge required to excel in tech-driven careers, making them not just job-ready but industry leaders of tomorrow.

We are not only limited to training the youth of the country but also provide Lab setups to various Colleges and Universities and Project solutions to other companies for the domains like **AI-ML, IOT , ROBOTICS AND CLOUD**.

NLP, Deep Learning & Generative AI		
Duration – 75 Hours		
1	**Introduction (4 Hours)**	• Introduction with AI & Machine Learning • Data Science V/s Data Engineering V/s Data Analysis vs AI • Use of Data in the world of Data Science and machine learning • Connecting with Upflairs Community • Basic Linux/Windows Commands • Setting Up GitHub & Google Colab/Kaggle
	Mini Project (2Hours)	***Project 1:*** • ***CLI Based Chat Application***
2	**Git & Github (4 hours)**	• What is GitHub? • Understanding repositories, commits, and branches. • Creating a GitHub account. • Creating a new repository. • Cloning a repository. • Making commits. • Pushing changes to GitHub. • Pulling changes from a repository. • Creating and switching branches. • Merging branches. • Resolving merge conflicts. • Forking a repository. • Creating pull requests. • Reviewing and merging pull requests.
3	**Deep learning (6 Hours)**	• Concept of Deep Learning & Neural Network • What is ANN? • The basic terminology – Layers, weights, biases, activation functions, losses, optimizers, learning rate • The Concept of Forward & Backward Propagation • Using Keras Library for ANN • Building and Compiling Sequential Neural Network Module
4	**Image Processing (8 Hours)**	• What is image processing? • Applications of image processing • Image Representation: Pixels and bit-depth, Grayscale vs. Color Images. • Introduction to OpenCV, PIL (Pillow) • Loading, displaying, and saving images. • Image attributes (size, shape, and data type) • Scaling, translation, rotation, and affine transformation. • Flipping, resizing, and cropping images. • RGB to Grayscale. • Concept of histograms. • Histogram equalization and its applications. • Erosion, Dilation, • object shape filtering, object border shape detection.

5	**Computer Vision with CNN (10 Hours)**	• what is computer vision • Real-world use cases of computer vision. • Understanding CNN working process • Filters, activation function, pooling layers, • flatten layers, CNN architecture, image feature extraction, • CNN architecture implementation. • Image classification. • Transfer learning techniques • Object detection: Localization using Yolo.
	Mini Project (12 Hours)	***Project2:*** • ***Dog cat image classification*** • ***image search project*** • ***Sign language detection*** • ***Leaf Disease detection yolo*** • ***Flower image classification***
6	**Natural Language Processing (4 Hours)**	• What is NLP? Linguistic to Natural Language! • NLTK in Python for Text Processing • Text to Speech and Speech to Text Modules in Python • Optical character recognition (OCR): Text recognition. • Generating Word Clouds
7	**Natural Language Processing with RNN (12 Hours)**	• What is RNN • How RNN is different from ANN • Types of RNN • Text cleaning steps, stemming and lemmatization, tokenization, stop words, pos tagging. Bag of words, tfidf, embedding layer. • The Concept of Long-Short-Term Memory (LSTM) • LSTM-based Neural Networks for Future Prediction!! • GRU.
8	**Generative AI (4 Hours)**	• What is Generative AI? • Real-world application of Generative AI. • Encoder and Decoder Based Architecture understanding, • Building Language Translator using encoder and decoder-based architecture using LSTM • What are LLM models? • How the LLM model works, • Learn about the LLM model working process, and generate the data. • Transformers Architecture understanding • BERT and GPT model architecture • NER extraction and text summarization using a pre-trained model • Named entity recognition (NER extraction)
	Minor project (11 Hours)	***Project 3:*** • ***Stock price prediction*** • ***Language translation*** • ***NER extraction and Text summarization using pre-trained model*** • ***Text generation Ai model from scratch***

Glance

UpFlairs

NIT, Jalandhar

Delhi Technical University

NIFFT, Ranchi

Banasthali Vidyapith

JECRC, Jaipur

SKIT,Jaipur

Arya College, Jaipur

Poornima College,Jaipur

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